

No. of Functional Features Included in the Solution

Date	25 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Home Page	Landing page introducing OptiCrop with navigation	Frontend (HTML/CSS)	Done	
2	About Page	Detailed project description and objectives	Frontend (HTML/CSS)	Done	
3	FindYourCrop	Interactive form for entering agricultural parameters	Frontend (HTML/CSS)	Done	
4	Previous Crops History	View and delete past predictions	Frontend & Backend	Done	
5	Crop Prediction	ML model predicts best crop based on user input	Backend (Flask + Pickle)	Done	
6	Data Processing	Handles missing values, outliers, feature scaling	ML Pipeline (Jupyter)	Done	
7	K-Means Clustering	Groups similar agricultural patterns with Elbow Method	ML Pipeline	Done	
8	Logistic Regression	Classifies crops using environmental parameters	ML Pipeline	Done	
9	Model Evaluation	Accuracy, Precision, Recall, F1, Confusion Matrix metrics	ML Pipeline	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	9
Total Features Implemented	9
Core / Must-Have Features	7
Additional / Nice-to-Have Features	2
Features Tested & Verified	9

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Home Page, About Page, FindYourCrop Page, Previous Crops Page
2	Backend / Logic	3	Crop Prediction API, Input Validation, History Management
3	Database / Storage	2	CSV dataset loading, JSON history, Model persistence (Pickle)
4	API / Integration	1	Flask prediction endpoint (/predict POST)
5	Security / Authentication	1	Input validation and range checking